

IoT Devices

An IoT (Internet of Things) device is a physical object embedded with sensors, software, and network connectivity to collect and exchange data over the internet or other networks



Smart Home: Smart speakers, thermostats, cameras, connected appliances.



Wearables: Fitness trackers, smartwatches monitoring health metrics.



Industrial (IIoT): IoT devices machinery for performance monitoring, asset trackers.



Automotive: Connected cars with GPS and diagnostic features.



Agriculture: Soil moisture sensors for precision farming.



Sensors & Actuators: They use sensors (e.g., temperature, motion) to gather data from the environment and actuators to perform actions.



Connectivity: They connect to networks (Wi-Fi, cellular, Bluetooth) to send and receive data, requiring an IP address to function.



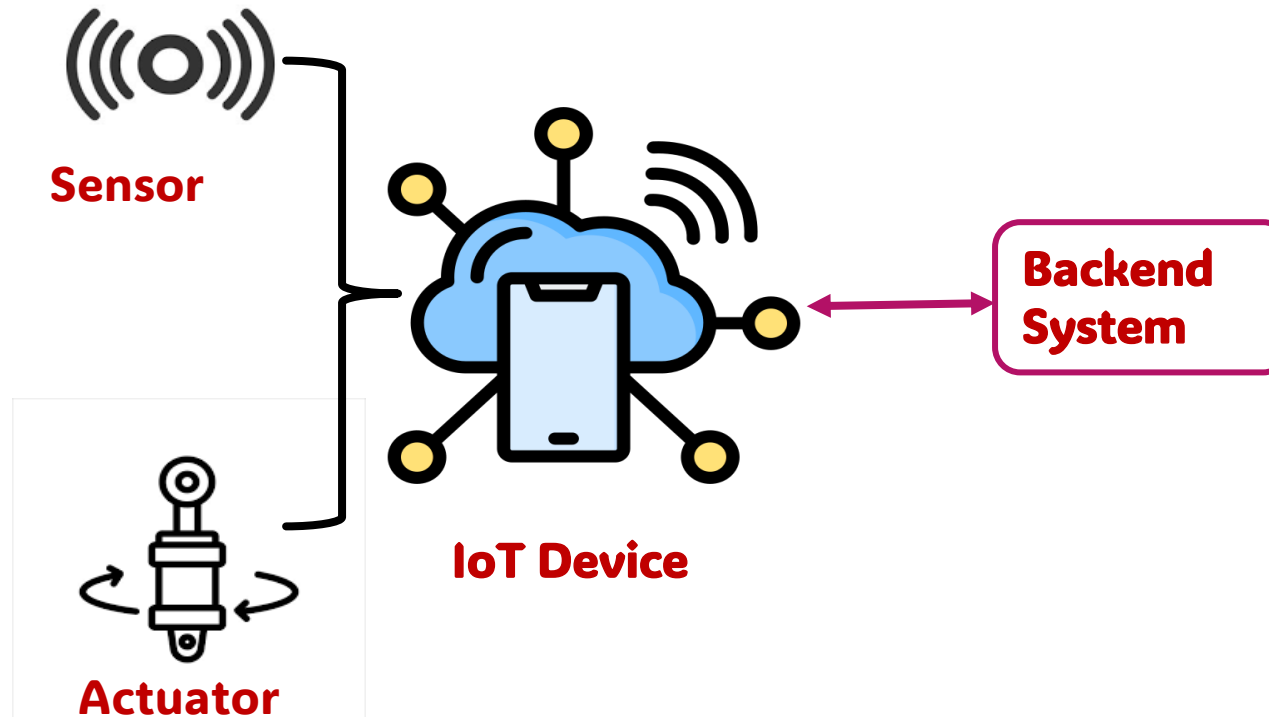
Processing Power: They have embedded CPUs and firmware to process data locally or send it for cloud analysis.



Data Exchange: They communicate with other devices, platforms, or apps, often managed by software, enabling automation and remote control.

IoT Devices

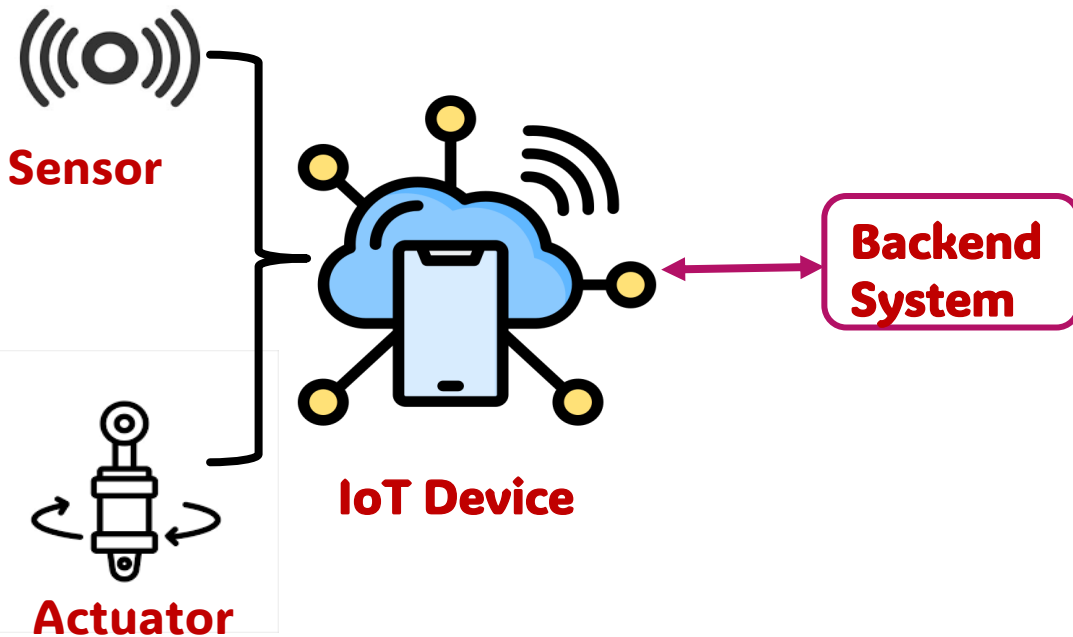
IoT is fundamentally about collecting information and then processing that information to make informed decisions



Device is essentially a bridge that integrates the sensor, the actuator, and the backend system.

IoT Devices

Processing power of Device



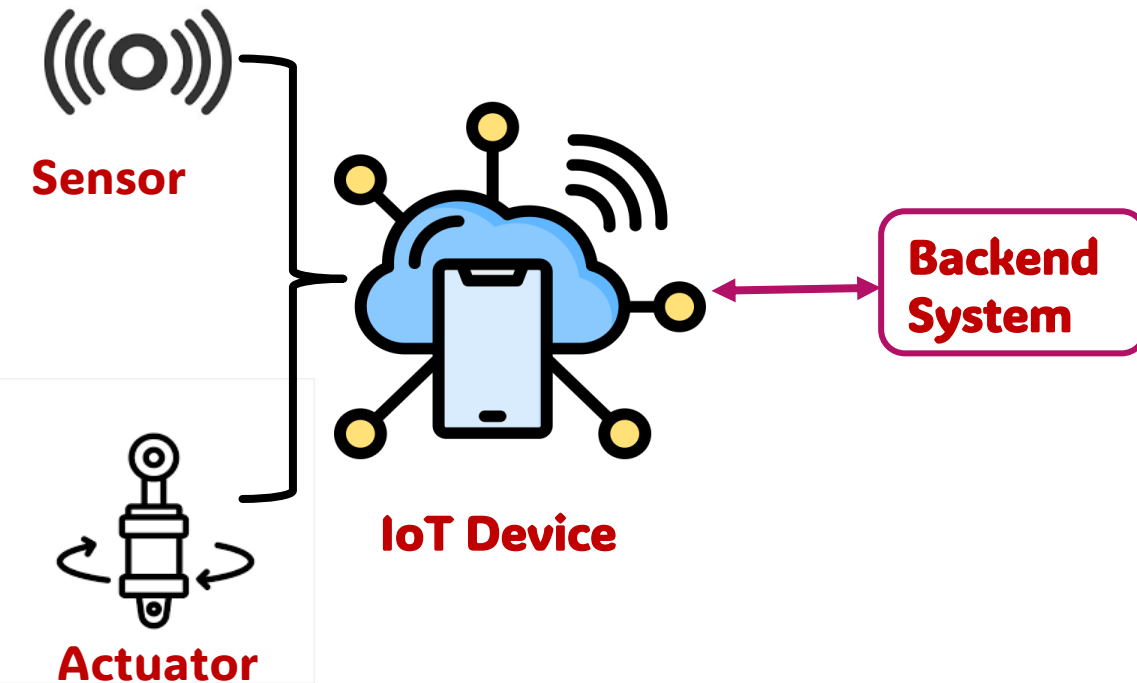
it would have abilities like Signal conditioning, A/D Converter, Processor, Communication.

Converts sensor data to appropriate format and sends the data for upstream processing/storage.

Receives control commands from upstream and triggers the actuators, processor, Local Transmission

IoT Devices

Additional feature



Support for **Local Display** and **Audio** - Processor, Display Port

Local Storage - RAM, SSD, extended memory

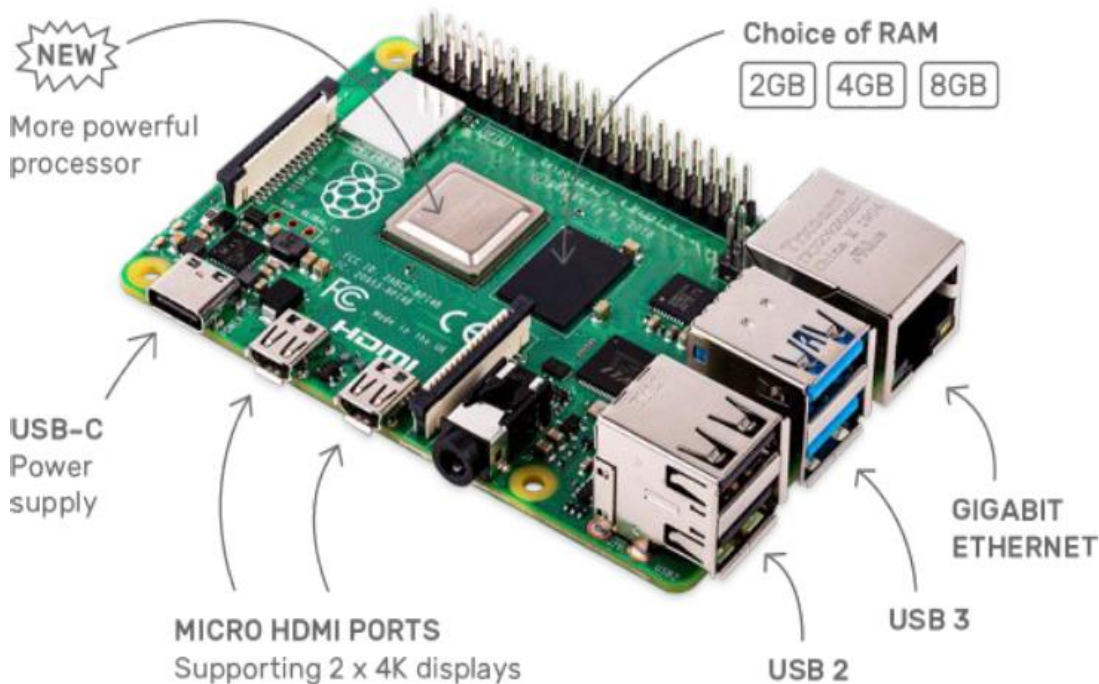
Powers to - Active Sensors, Actuators, Processor, Network subsystem, Display, Data converters, Memory

May have many other things like LED, interfaces like USB

IoT Devices

General Purpose Devices for Learning and Testing

Devices - Raspberry Pi 4 Anatomy



General Purpose Devices are available in market for Learning and Testing.

Ready to use devices and Need to check how much it suites to the use case

Provide basic display, processing power , n/w capabilities to transfer information to cloud, edge devices or local computers.

Ability to connect various sensors and actuators

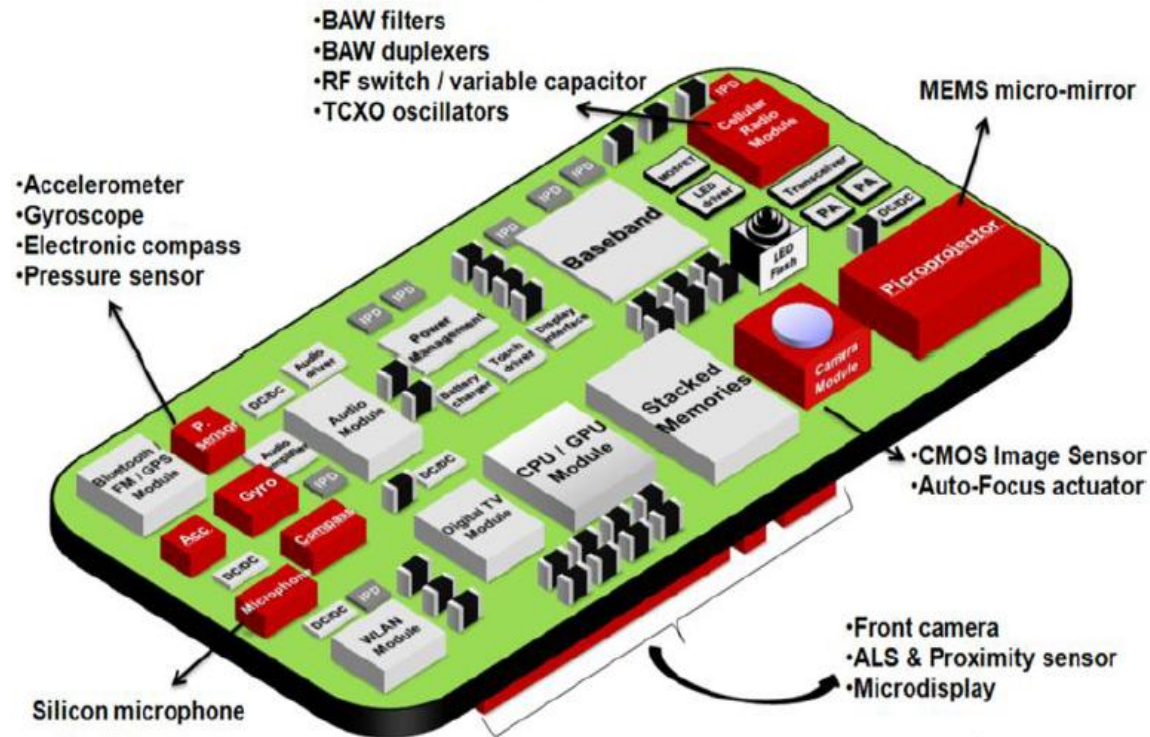
Have IDE, toolkit, SDK, libraries , drivers

Examples :-

- ESP32 series , u-blox, NXP Rapid IoT kit,
- Raspberry Pi 4,
- Arduino Uno/Leonardo

IoT Devices

Devices - Typical Smartphone Anatomy



Purpose-built Devices - Production Use

- Embedded sensors and actuators for a specific purpose
- Targeted processing and power requirements
- Specific network support based on topology, bandwidth and power
- Secure data storage and transmission, physical tamper-proofing
- Environment Safety (dust/water/hazards) - Ingress Protection (eg. IP68)
- Examples :
 - InnoSent - Smart Home and Smart Building Devices
 - DSEgenset - Sensor/Devices for Power Generator control
 - Google Nest Audio, Amazon Alexa, Your Smartphone

Device Processing Capabilities

This is deep engineering area. Below are few pointers to consider

Device hardware

How to Estimate?

• Processing speed

- Sensor senses at 50ms
- Processor takes 100ms (read, transform, store/transfer)
- Sensor capability unutilized

• Power consumption limits the maximum processing power

• Typical Processor Types few examples

- 8 bit, 2-40 MHz
- 16 bit, 2-250 MHz
- 32 bit, 2-1000 MHz on a single core
- 32bit, Multicore

Device Software

What do you run?

• Smaller devices

- Basic system management support
- Logic written in assembly or C

• Larger Devices

- FreeRTOS - Popular real time open-source kernel/OS
- Ported on numerous platforms
- MbedOS - OS/Platform by ARM with support for C/C++
- RTLinux - Real-time Linux variant

Device Networking Capability

Networking interface and protocol is a key decision

• Core considerations

- Power -Peak limit, long term range
- Protocols : TCP/UDP, IP assignment
- Wired/Wireless
- Distance to the receiver
- Bandwidth available at the source
- Data size per transfer
- Security & compliance requirements
- Overall topology of the device/sensor

• We'll dig deeper into Networking, in the next week session